

A noncommuting singular-commutator class satisfying every matrix rearrangement inequality

Candidate counterexample packet for arXiv:1904.05239

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Status: candidate full counterexample to necessity, likely valid, subject to human review. There are noncommuting positive definite 3×3 matrices A, B for which $\ker(AB - BA) \neq \{0\}$, but every rearrangement inequality in the source holds, globally and also for $(\text{Id} + \varepsilon A, \text{Id} + \varepsilon B)$ for every $\varepsilon > 0$. The same argument gives a simple sufficient condition: a common orthogonal reducing decomposition into blocks of dimension at most two.

1 Source question

Alaifari, Cheng, Pierce, and Steinerberger prove that for positive semidefinite 2×2 matrices C, D , every word satisfies

$$\|C^{m_1} D^{n_1} \dots C^{m_s} D^{n_s}\| \leq \|C^m D^n\|, \quad m = \sum_i m_i, \quad n = \sum_i n_i. \quad (1)$$

They also prove a near-identity theorem in arbitrary dimension under the condition $\ker(AB - BA) = \{0\}$ [1, Theorems 1 and 2]. Immediately after Theorem 2, on page 3, they write:

“We do not know whether the condition $\ker(AB - BA) = \emptyset$ is necessary but are inclined to think that it may not be.”

Here and below the printed $\emptyset$ is read as the zero subspace. They also ask for simple sufficient conditions other than commutativity. The source page is reproduced in Appendix A.

The literal commuting example already shows that the condition is not necessary, and the source explicitly notes this trivial case. The result below removes that loophole: its matrices do not commute, their commutator has a nonzero kernel, and the conclusion holds for every word and every positive perturbation parameter.

Idea of the proof

The source’s 2×2 theorem can be assembled blockwise. Products and powers respect a common orthogonal direct-sum decomposition, and the spectral norm of a direct sum is the maximum of the block norms. Hence every pair whose common reducing blocks have size at most two satisfies all word inequalities. Taking one genuinely noncommuting 2×2 block and adjoining a scalar block makes the full pair noncommuting while forcing a kernel in the commutator. Choosing the scalar value larger than both block norms even puts the top eigenspace of every $mA + nB$ inside that kernel.

2 Block stability

For nonnegative integers m_i, n_i as in the source, write

$$W(C, D) = C^{m_1} D^{n_1} \dots C^{m_s} D^{n_s}.$$

Lemma 1 (direct-sum closure). *Suppose*

$$C = \bigoplus_{j=1}^r C_j, \quad D = \bigoplus_{j=1}^r D_j$$

are positive semidefinite matrices on a common orthogonal decomposition. If

$$\|W(C_j, D_j)\| \leq \|C_j^m D_j^n\|$$

for every j , then

$$\|W(C, D)\| \leq \|C^m D^n\|.$$

Proof. Functional calculus and multiplication are blockwise, so

$$W(C, D) = \bigoplus_j W(C_j, D_j), \quad C^m D^n = \bigoplus_j C_j^m D_j^n.$$

The spectral norm of an orthogonal direct sum is the maximum of the block norms. Therefore

$$\|W(C, D)\| = \max_j \|W(C_j, D_j)\| \leq \max_j \|C_j^m D_j^n\| = \|C^m D^n\|.$$

□

Corollary 1 (a simple noncommutative sufficient condition). *If two real symmetric positive semidefinite matrices have a common orthogonal reducing decomposition into blocks of dimension at most two, then they satisfy every rearrangement inequality (1).*

Proof. On a one-dimensional block the two matrices commute, so equality holds. On a two-dimensional block, inequality (1) is Theorem 1 of the source. Apply Lemma 1. □

3 Explicit counterexample to necessity

Theorem 1. *Let*

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

Then:

1. A and B are positive definite and do not commute;
2. $\ker(AB - BA) = \text{span}\{e_3\}$;
3. for every admissible word,

$$\|W(A, B)\| \leq \|A^m B^n\|;$$

4. for every $\varepsilon > 0$ and every admissible word,

$$\|W(\text{Id} + \varepsilon A, \text{Id} + \varepsilon B)\| \leq \|(\text{Id} + \varepsilon A)^m (\text{Id} + \varepsilon B)^n\|.$$

Thus the commutator-kernel condition in the source’s near-identity theorem is not necessary, even among noncommuting positive definite 3×3 pairs.

Proof. Set

$$A_0 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad B_0 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The eigenvalues of A_0 are 2, 1, and those of B_0 are 3, 1; hence $A = A_0 \oplus [4]$ and $B = B_0 \oplus [4]$ are positive definite. Direct multiplication gives

$$AB - BA = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus A, B do not commute and the kernel of their commutator is exactly $\text{span}\{e_3\}$.

The pair has a common orthogonal reducing decomposition $\mathbb{R}^3 = \mathbb{R}^2 \oplus \mathbb{R}e_3$ with block dimensions two and one. Part (3) follows from Corollary 1. For part (4), observe that

$$\text{Id} + \varepsilon A = (I_2 + \varepsilon A_0) \oplus [1 + 4\varepsilon], \quad \text{Id} + \varepsilon B = (I_2 + \varepsilon B_0) \oplus [1 + 4\varepsilon].$$

Both two-dimensional blocks are positive definite for $\varepsilon > 0$, so the same corollary applies. Equivalently, on the scalar block the two sides are equal to $(1 + 4\varepsilon)^{m+n}$, while the source’s 2×2 theorem controls the other block; taking the maximum preserves the inequality. $\square$

There is also a robustness check against a natural weakening of the printed hypothesis. If V_1 denotes the top eigenspace of $mA + nB$ and $m, n \geq 1$, then on e_3 the eigenvalue is $4(m + n)$, whereas

$$\|mA_0 + nB_0\| \leq 2m + 3n < 4(m + n).$$

Consequently $V_1 = \text{span}\{e_3\} \subset \ker(AB - BA)$. Thus the example also violates the weaker exclusion $V_1 \cap \ker(AB - BA) = \{0\}$ while retaining all rearrangement inequalities.

4 Verification, scope, and novelty

The formal proof uses only block multiplication, the maximum formula for the spectral norm of an orthogonal direct sum, and Theorem 1 of the source. The matrix arithmetic and random finite word tests are independently checked by `code/verify_example.py`; those computations are sanity checks, not part of the proof.

This answers the source’s necessity question in the negative under its nontrivial, noncommuting interpretation and supplies the adjacent requested simple sufficient condition. It does not address the source’s broad measure-theoretic question about generic admissibility, nor does it classify all pairs satisfying every word inequality.

The bounded novelty check on 11 August 2026 searched the run registry and its solution, attempt, and proof-gap indexes for the arXiv id, title, commutator kernel, and block-rearrangement phrases. Exact and close-phrase primary-source web searches found the source preprint, its published record, and author publication pages, but no later paper resolving this necessity question. Because the argument is elementary, this is not an exhaustive priority claim.

Human-review recommendation. Verify that the authors intended “necessary” to exclude not only commuting pairs but also reducible noncommuting pairs. Under that natural reading, Theorem 1 is decisive; if irreducibility was intended but unstated, the irreducible variant remains open.

A Source-page evidence

References

- [1] R. Alaifari, X. Cheng, L. B. Pierce, and S. Steinerberger, *On Matrix Rearrangement Inequalities*, Proc. Amer. Math. Soc. 148 (2020), 1835–1848; arXiv:1904.05239v2 [math.FA].

Theorem 1 (General Rearrangement Inequality for 2×2 Matrices). *Let A, B be symmetric positive semidefinite matrices of size 2×2 and let $m_i, n_i \in \mathbb{N}$ (possibly allowing $m_1 = 0$ or $n_s = 0$). Then*

$$\|A^{m_1} B^{n_1} A^{m_2} B^{n_2} \dots A^{m_s} B^{n_s}\| \leq \|A^m B^n\|,$$

where $m = \sum_i m_i$ and $n = \sum_i n_i$.

In light of Drury's results, there is no hope for such general inequalities in higher dimensions. Nonetheless, one could wonder whether there is hope that, given any word $(m_1, n_1, \dots, m_s, n_s)$, a rearrangement inequality should hold for *some* (or maybe even *most*) pairs of $N \times N$ matrices (A, B) . This is the motivation for our second result, which states that given any word, the rearrangement inequality is generically true for $N \times N$ matrices in a sufficiently small neighborhood of the identity, for all $N \geq 2$.

Theorem 2 (General Rearrangement close to the Identity, arbitrary dimension). *Let A, B be symmetric positive semidefinite matrices and let $m_i, n_i \in \mathbb{N}$ (possibly allowing $m_1 = 0$ or $n_s = 0$). If $\ker(AB - BA) = \emptyset$, then there exists $\varepsilon_0 = \varepsilon_0(A, B, m, n) > 0$ such that for all $0 < \varepsilon < \varepsilon_0$*

$$\|(\text{Id} + \varepsilon A)^{m_1} (\text{Id} + \varepsilon B)^{n_1} \dots (\text{Id} + \varepsilon A)^{m_s} (\text{Id} + \varepsilon B)^{n_s}\| \leq \|(\text{Id} + \varepsilon A)^m (\text{Id} + \varepsilon B)^n\|,$$

where $m = \sum_i m_i$ and $n = \sum_i n_i$.

Thus given any fixed word, this provides a codimension 1 family of (A, B) among all relevant pairs of $N \times N$ matrices in the neighborhood of the identity, which satisfy the rearrangement inequality for that word. We do not know whether the condition $\ker(AB - BA) = \emptyset$ is necessary but are inclined to think that it may not be.

There are many other natural questions that come to mind. The rearrangement inequalities are invariant under multiplication with constants, which allows us to compactify the set of matrices: are such inequalities *generically* true (in, say, the sense that the measure of admissible matrices approaches full measure as the length of the inequality, or the number s , increases)? Another question could be to determine other simple conditions on the matrices (other than assuming that they commute) that would imply the desired rearrangement inequalities hold.

2. PROOF OF THEOREM 1

Our proof uses three different ingredients. The first ingredient is Corollary 4.4 in a paper of Ando, Hiai & Okubo [2] which states the following: let C, D be symmetric positive semidefinite matrices of size 2×2 and for $i = 1, \dots, k$, let $p_i, q_i \geq 0$ satisfy

$$p_1 + \dots + p_k = 1 = q_1 + \dots + q_k;$$

then

$$(2) \quad \text{tr}(C^{p_1} D^{q_1} \dots C^{p_k} D^{q_k}) \leq \text{tr}(CD).$$

We remark that Ando, Hiai & Okubo [2] were motivated by the question whether such a trace inequality might be true in general: they establish the result for general positive semidefinite matrices that have at most two distinct eigenvalues. Plevnik [24] recently constructed an example showing that (2) can fail for 3×3 matrices.

The second ingredient is the invariance of trace with respect to cyclic permutations, i.e.

$$(3) \quad \text{tr}(A_1 A_2 \dots A_n) = \text{tr}(A_2 A_3 \dots A_{n-1} A_n A_1).$$

The third ingredient is the basic equation

$$(4) \quad \|A\|^2 = \max_{\|x\|=1} \langle Ax, Ax \rangle = \max_{\|x\|=1} \langle A^T A x, x \rangle = \lambda_{\max}(A^T A),$$

where $\lambda_{\max}$ denotes the largest eigenvalue of a matrix.

Let A, B be symmetric positive semidefinite matrices of size 2×2 . Consider now a general word

$$(5) \quad W_{m,n}(A, B) := A^{m_1} B^{n_1} \dots A^{m_s} B^{n_s}$$

Figure 1: Page 3 of arXiv:1904.05239v2. Theorem 1 gives the 2×2 input; the text following Theorem 2 asks whether the commutator-kernel condition is necessary and asks for other simple sufficient conditions.